

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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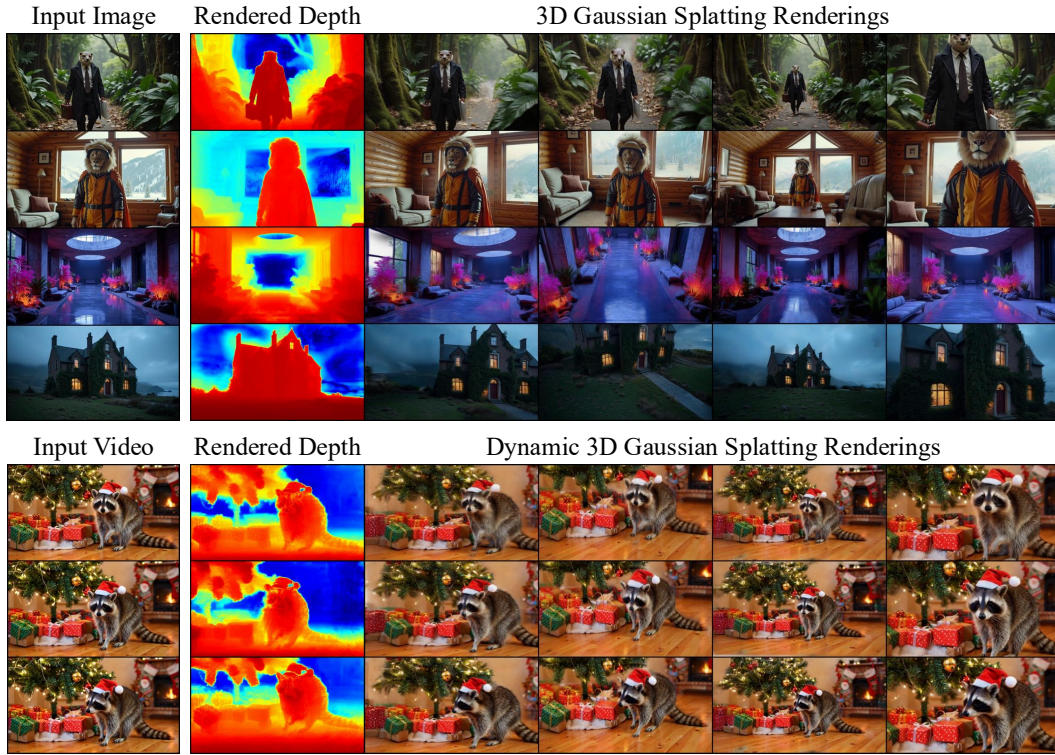


Figure 1: Feed-Forward 3D and 4D Scene Generation. From a single image (top), *Lyra* uses self-distillation from a video diffusion model to infer a 3D Gaussian Splatting (3DGS) representation in one feed-forward pass, without real-world multi-view training data. From a video input (bottom), *Lyra* extends the same framework to infer a dynamic 3DGS, exposing interactive control over both time (rows) and viewpoint (columns).

ABSTRACT

Generating virtual environments is valuable for gaming, robotics, autonomous driving, industrial AI, and other settings that need controllable worlds for simulation and interaction. The central bottleneck is that existing learning-based 3D reconstruction methods typically require captured real-world multi-view data, which is expensive, difficult to scale, and often unavailable. Video diffusion models offer a complementary source of supervision: they can imagine diverse photorealistic videos, but their outputs remain 2D and therefore cannot directly support robotic navigation, interaction, or real-time 3D rendering. This paper contributes a self-distillation framework that converts the implicit 3D knowledge of a camera-controlled video diffusion model into an explicit 3D Gaussian Splatting (3DGS) representation, removing the need for real multi-view training captures. Our key design is to augment the standard RGB decoder with a latent-space 3DGS decoder and train this decoder from the RGB outputs of the video model. Because supervision comes from synthetic videos generated by the diffusion model itself, the 3DGS decoder can be trained entirely from generated data while preserving the

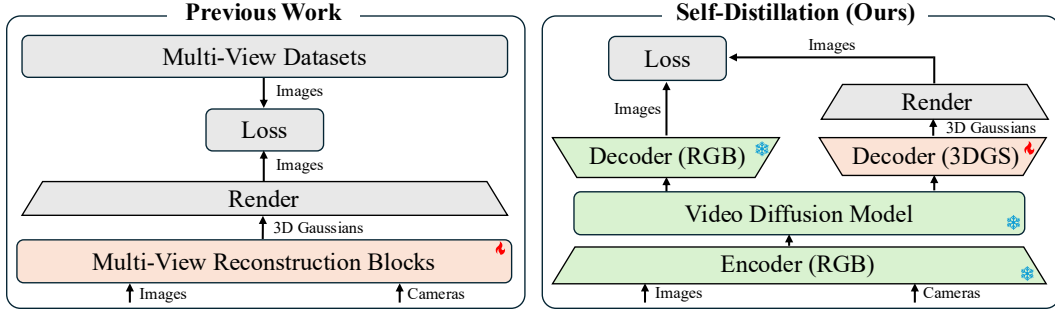


Figure 2: **Self-distillation framework of *Lyra*.** Previous work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks on real-world datasets with limited diversity (Zhou et al., 2018; Ling et al., 2024b). Our contribution (right) replaces that dependency with self-distillation for generative 3D scene reconstruction: a pre-trained camera-controlled video diffusion model and its RGB decoder output serve as the teacher, while renderings from the 3DGS decoder serve as the student. Because the teacher is pre-trained on diverse 2D video data, the student receives diverse multi-view supervision without real multi-view captures.

diversity of the video prior. At inference time, the resulting model synthesizes 3D scenes from either a text prompt or a single image and supports real-time rendering. The same contribution structure extends to dynamic 3D scene generation from a monocular input video by producing time-conditioned 3DGS representations. Experiments show that this self-distilled generative reconstruction framework achieves state-of-the-art performance for both static and dynamic 3D scene generation.

1 INTRODUCTION

Creating high-quality 3D environments at scale remains a core challenge for computer vision and graphics. The value of such environments is clearest in applications such as film-making, VR/AR, and physical AI with closed-loop simulation, where a scene must be represented explicitly enough to support real-time rendering, physical interaction, and consistent multi-view synthesis.

Neural 3D reconstruction methods (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) provide one route to these explicit representations, but their contribution is bounded by the data they require. They depend on accurate camera poses and high-quality posed images, which makes scaling difficult; for dynamic scenes, this dependence often grows into synchronized multi-camera capture (Li et al., 2022). They are also limited to observed content and cannot naturally extrapolate beyond the captured views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) reduce per-scene optimization, but they expose a second bottleneck: diverse, large-scale 3D training data are scarce, and this scarcity weakens out-of-domain generalization.

Video diffusion models (Cosmos, 2025; Wan, 2025) suggest a different source of supervision. Trained on internet-scale video, they generalize across diverse environments and implicitly observe many camera trajectories, from handheld motion to cinematic pans and drone footage. Unlike reconstruction-only systems (Charatan et al., 2024), they can hallucinate plausible unobserved content. Their limitation, however, is exactly where our paper intervenes: the generated output is a sequence of 2D frames rather than an explicit 3D representation, which restricts use in simulation and downstream tasks requiring geometric consistency, long-term coherence, and physical interaction. Recent camera-controlled video models (Wang et al., 2024e; Ren et al., 2025) show that video diffusion models can be guided by posed trajectories using relatively small posed datasets, making them a practical teacher for explicit 3D reconstruction.

We introduce *Lyra* as a generative 3D scene reconstruction framework that makes this teacher signal explicit. Crucially, *Lyra* contributes a self-distillation procedure (Fig. 2) in which a 3D Gaussian Splatting (3DGS) decoder operates directly in the latent space of a video diffusion model and learns from the model’s own RGB decoder. Given a single input image or video, we sample camera trajectories, denoise the corresponding video latents, and decode them through two branches: the standard RGB branch produces teacher videos, while the 3DGS branch produces an explicit scene representation. Multi-trajectory supervision then trains the 3DGS decoder to fuse information across viewpoints and reduce long-term and multi-view inconsistencies.

This framing makes three contributions visible throughout the paper. **First**, self-distillation turns camera-controlled video generation into scalable supervision for 3D reconstruction, avoiding real-world multi-view training captures. **Second**, latent-space reconstruction lets the model aggregate many generated views without the memory cost of pixel-space feed-forward reconstruction. **Third**, the output is an explicit 3DGS scene, so inference produces real-time-renderable 3D Gaussian scenes from monocular input *without requiring any additional optimization or post-processing*. Together, these design choices bridge reconstruction and generation while preserving the downstream utility of explicit geometry.

We also extend the same contribution to dynamic 4D generation from monocular video. In this setting, the video model provides space-time teacher supervision, and the student learns time-conditioned 3DGS representations that support novel-view synthesis of dynamic scenes.

Overall, our work makes the following contributions:

- We introduce a self-distillation framework that trains a 3DGS student from a camera-controlled video diffusion teacher, enabling explicit 3D scene generation without real-world multi-view training data.
- We design a latent-space 3DGS decoder that fuses multiple generated video trajectories efficiently and outputs real-time-renderable 3D Gaussians in a single forward pass.
- We extend the framework to dynamic scenes by learning time-conditioned 3DGS representations from monocular video supervision.
- We demonstrate state-of-the-art static and dynamic 3D generation results, including improved quantitative performance and broad qualitative generalization across diverse generated scenes.

2 RELATED WORK

Multi-view image generation. Prior multi-view image generation work provides important context for our contribution because it separates view synthesis from explicit 3D reconstruction. Early methods (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) mainly target object-centric scenes without backgrounds, while later systems (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) scale multi-view image generation toward scenes. CAT3D (Gao et al., 2024b), for example, adapts a pre-trained image diffusion model to generate multi-view images from generated or real image inputs, then reconstructs those views into NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). Related space-time diffusion models extend this pipeline to dynamic 3D scenes from multiple viewpoints (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). These approaches demonstrate strong multi-view synthesis, but the explicit 3D representation still comes from a separate, expensive per-scene optimization stage. Our contribution instead targets direct amortized generation of an explicit 3D scene from text or a single image.

Camera-conditioned video generation. Camera-conditioned video generation is the enabling teacher for our method. MotionCtrl (Wang et al., 2024e) introduced camera control by conditioning pre-trained video models on camera poses, and follow-up work (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) uses Plücker coordinates for pixel-wise conditioning. Other methods (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) study training-free camera control, while SynCam-Master (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) show strong results from large-scale synthetic training data. These models contribute controllable 2D video synthesis, but their outputs are not grounded in a reusable 3D representation. Our work is orthogonal: we use GEN3C (Ren et al., 2025), a camera-controlled video diffusion model, as a self-distillation teacher that trains a feed-forward 3D reconstruction student and enables novel-view rendering *after* generation.

Feed-forward 3D models. Feed-forward 3D models motivate both the output representation and the scalability gap addressed by our latent decoder. Early approaches focus on object-centric scenes and directly predict NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) representations from text or an image. Scene-level systems (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) typically regress per-pixel 3D Gaussians from one or more input images, but they often generalize best within training distributions such as RealEstate10K (Zhou et al., 2018). Recent generative reconstruction methods narrow this gap: Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder for feed-forward 3D reconstruction from multi-view images, and Wonderland (Liang et al., 2025a) uses a camera-controlled video model to generate 3D Gaussians with a feed-forward network. In contrast,

our contribution is to train the 3D student through self-distillation from the camera-controlled video model *without requiring real-world multi-view data*, and to show that the same framework naturally extends to feed-forward 4D scene generation.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

The method contribution is to convert the implicit 3D knowledge of a camera-controlled video diffusion model into an explicit 3DGS decoder. We implement this conversion as a teacher–student system: the video diffusion model acts as the teacher by generating RGB videos, and the 3DGS decoder acts as the student by operating in the same latent space and producing renderable Gaussians, as illustrated in Fig. 2. This section first identifies the video diffusion backbone that supplies 3D-consistent supervision (Sec. 3.1), then describes the self-distillation objective for the 3DGS decoder (Sec. 3.2). We then explain the latent 3DGS decoder design (Sec. 4) and the minimal changes that extend the same contribution to dynamic 3D scenes (Sec. 5).

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because our student receives supervision only from the camera-controlled video diffusion model, the teacher must provide 3D-consistent video guidance. We therefore build on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model designed for controlled novel-view video generation. The following components are the aspects of GEN3C that matter for our self-distillation pipeline and its generated video latent $\mathbf{z}$.

Spatiotemporal 3D cache. The first useful property is structured 3D conditioning. GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos, where each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimation (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, where L denotes the number of frames (temporal length) and V denotes the number of camera views. This cache gives the teacher an explicit geometric scaffold before it hallucinates disoccluded content.

Rendering and structured guidance. The second useful property is that GEN3C exposes where generation must complete missing content. To use the cache, GEN3C renders each point cloud from arbitrary camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ denotes the rendering function that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane according to the camera pose $\mathbf{C}^t$ at time t . Given a sequence of camera poses $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache elements produces V videos of length L , which can be stacked into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The masks mark disoccluded regions that the video diffusion model should fill in, and the rendered RGB/mask pairs become the structured guidance used by the teacher.

Video variational autoencoder (VAE). The third useful property is a compact latent space in which our student can scale. Diffusion-based video generation models commonly use a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress videos for efficient training and inference. Given a RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ encodes the video into a latent space, i.e. $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$. The diffusion model is trained and evaluated in this latent space, and the final video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is decoded with a pre-trained VAE decoder $\mathcal{D}_{rgb}$. In this paper, we adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model. Specifically, the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, where the temporal compression factor is $\tau = 8$ and the spatial compression factor is $\sigma = 8$.

3.2 SELF-DISTILLATION

The self-distillation contribution is to treat generated videos as supervision rather than as final outputs. We train the 3DGS decoder $\mathcal{D}_s$ as a student, with a camera-controlled video diffusion model $\mathcal{V}$ serving as the teacher. To make this supervision broad, we curate diverse text prompts with large language models (OpenAI, 2025; Bai et al., 2023), generate images I with an image diffusion model (Black Forest Labs, 2024), and expand them into multi-view sequences using GEN3C (Ren et al., 2025).

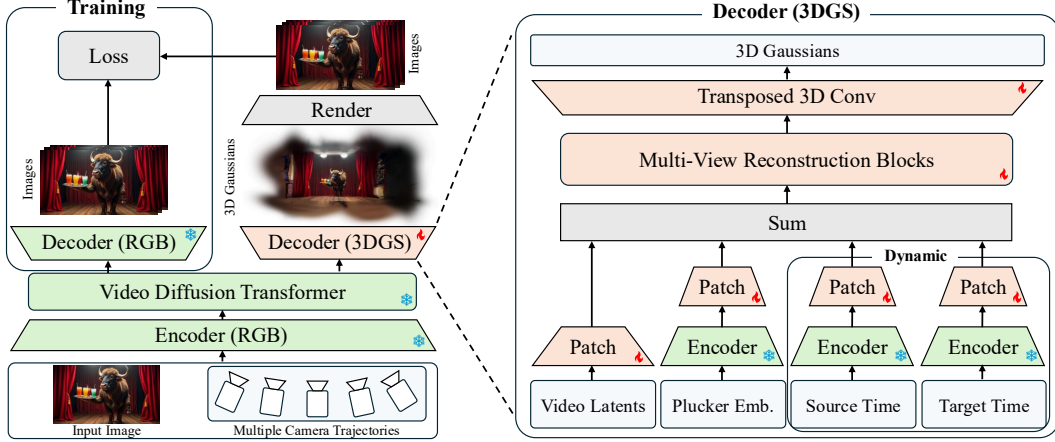


Figure 4: **3D Generative Reconstruction Framework.** The framework contributes a self-distillation path from a pre-trained camera-controlled video diffusion model (Ren et al., 2025) to an explicit 3DGS decoder. During training, renderings from generated 3DGS scenes are aligned with RGB-decoded generations from the frozen video model and autoencoder, so only the 3DGS decoder is learned. At inference time, the RGB decoder is no longer required because the 3DGS decoder directly outputs the renderable scene. Adding time conditioning to the same decoder extends the framework from static to dynamic 3D scene generation.

Teacher-student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ generates a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. We decode this same latent through two branches. The pre-trained RGB decoder $\mathcal{D}_{rgb}$ produces the teacher video $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ produces explicit 3D Gaussians $\mathbf{G}$. The student renderings are $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$, and they are supervised to match the teacher frames $\mathbf{I}_{\mathcal{D}_{rgb}}$. This loop is the mechanism that distills the video model’s latent 3D consistency into an explicit scene representation.

Multi-trajectory supervision. A single trajectory provides limited coverage, so our training signal explicitly emphasizes multi-view fusion. For each input image, we sample $V = 6$ camera trajectories, as shown in Fig. 3. For each trajectory v , we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ and $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through the video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ is trained to fuse all $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$ while filling disoccluded regions. This design makes the practical benefit of self-distillation explicit: the 3DGS decoder is trained entirely from synthetic supervision provided by $\mathcal{V}$, latent-space aggregation keeps many trajectories tractable, and the decoder $\mathcal{D}_s$ produces Gaussian output $\mathbf{G}$ that supports geometric consistency, downstream simulation, and real-time rendering.

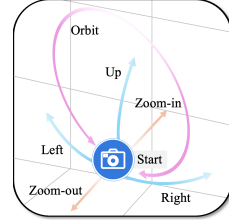


Figure 3: Six sampled camera trajectories provide the multi-view coverage used for self-distillation.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

The reconstruction contribution is a feed-forward 3DGS decoder $\mathcal{D}_s$, implemented in the same latent path, that turns synthesized multi-view video latents from our video model $\mathcal{V}$ into an explicit 3D representation renderable from arbitrary viewpoints. We use 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) because it is explicit and fast to render. Fig. 4 (right) visualizes the decoder architecture.

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. The key scaling choice is to reconstruct from video latents instead of pixels. Existing feed-forward reconstruction frameworks generate 3D Gaussians for each pixel, which limits the number and resolution of input views they can process. For example,

GS-LRM (Zhang et al., 2024e) operates on 2–4 images at 512×512 resolution, and AnySplat (Jiang et al., 2025a) is trained with 24 images at 448×448 resolution. Our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution, far beyond the capacity of pixel-space methods and GPU memory limits. The bottleneck is the attention over visual tokens, whose memory and compute scale with the number of pixels. We therefore perform reconstruction directly in the compressed video latent space produced by the camera-controlled video diffusion model, with multi-view video latents denoted as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. Using $\mathcal{E}$ -aligned inputs, the 3DGS decoder $\mathcal{D}_s$ maps latent inputs into reconstruction-block features and outputs per-pixel 3D Gaussian attributes $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Its inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both in latent dimensions $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each component is patchified to the hidden dimension of the main reconstruction blocks; no additional visual encoder is needed, only a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) that converts video latents into flattened reconstruction tokens. The summed inputs pass through reconstruction blocks following Long-LRM (Ziwen et al., 2024): one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers. We repeat this block twice to obtain 16 layers with 512 hidden dimensions. A final transposed 3D convolution maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is expressed as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. Camera information enters the decoder through latent Plücker embeddings aligned with the video latents. We first compute raw Plücker embeddings from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. We then reuse the RGB encoder $\mathcal{E}$ from the pre-trained VAE to encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins separately. Concatenating these latents along the channel dimension gives $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$, and a lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of 2 so it matches video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. The primary distillation signal is image-space agreement between student renderings and teacher frames. We split this reconstruction loss into a Mean Squared Error (MSE) loss $\mathcal{L}_{mse}$ and an LPIPS loss $\mathcal{L}_{lips}$ (Zhang et al., 2018) using VGG (Simonyan & Zisserman, 2014) as the feature extractor.

Depth supervision. RGB supervision alone can produce flattened geometry, so we add depth supervision to make the explicit 3D contribution more stable. We supervise rendered depth maps using consistent video depth estimated by the off-the-shelf system ViPE (Huang et al., 2025) and the scale-invariant depth loss $\mathcal{L}_{depth}$ from Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. To keep the generated representation compact, we follow Long-LRM (Ziwen et al., 2024), apply L1 regularization to opacity $\mathcal{L}_{opacity}$, and remove the Gaussians with the lowest 80% opacity.

Total loss. The final objective combines the image, perceptual, depth, and opacity terms with weight factors λ_i :

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lips} \mathcal{L}_{lips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

The dynamic-scene contribution is that the same teacher–student recipe extends from static 3D to monocular-video 4D generation with only targeted architectural and data changes.

Self-distillation for dynamic scenes. The dynamic setup preserves the static design while replacing the single image with a single video of length L and corresponding camera poses. The video model generates multi-view video latents that share the same underlying motion, giving the student space–time supervision. For video inputs, we follow the static data protocol: sample diverse text prompts with large language models, generate videos using video diffusion models (Cosmos, 2025; Wan,

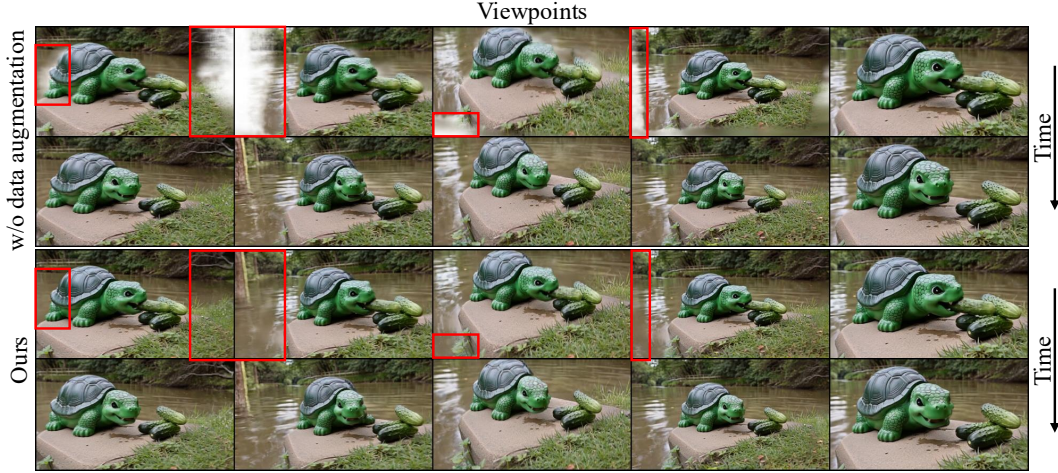


Figure 5: **Dynamic data augmentation.** This ablation highlights the contribution of motion-reversed supervision for time-aware 3DGS decoding. Naive supervision produces low-opacity artifacts at early timesteps from extreme poses because parts of the scene are not covered by the supervision signal. Adding a motion-reversed video gives each timestep broader spatial coverage and prevents these early-timestep opacity artifacts.

2025), and annotate the generated videos with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. To make the explicit representation time dependent, we follow the bullet-time design of Liang et al. (2025b) and define a dynamic 3DGS decoder $\mathcal{D}_s$; this $\mathcal{D}_s$ generates 3D Gaussians for all video frames. The architecture closely follows $\mathcal{D}_s$, with the key addition that $\mathcal{D}_s$ receives encoded source and target time embeddings $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The unencoded source times are assigned to each input frame, forming $\mathbf{T}^{\text{src, raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the target time is $\mathbf{T}^{\text{tgt, raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are spatially repeated, augmented with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, and encoded with the RGB encoder $\mathcal{E}$ to latent dimensions $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, a target time step is randomly sampled, and the corresponding 3D Gaussians are decoded and supervised across all trajectories generated with dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$ with the patchification layers of $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ initialized to zeros. $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are directly added to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. Dynamic supervision introduces a contribution-specific challenge: unlike static scenes, where frames from different timesteps can supervise the same 3DGS renderings, dynamic scenes require timestamp-matched supervision because the 3DGS changes over time. This restriction can lead to a trivial solution in which the 3DGS at a given timestep ignores frames from other timesteps (see Fig. 5).

We address this failure mode with dynamic data augmentation that balances near and far supervision at every timestep. Early frames are naturally associated with viewpoints near the input image, while later frames correspond to more distant viewpoints, so we add paired supervision views for each timestep. Concretely, we reverse the input video in frame order and feed it into the video model $\mathcal{V}$, which produces six additional multi-view sequences. After reversing these sequences back to the original motion order, we obtain six trajectories that progress inward from far viewpoints toward the input. Combined with the original six outward trajectories, this yields 12 supervision views per timestep. This augmentation is used only during training: the reversed trajectories provide extra supervision that prevents collapse from pruning artifacts and are not required at inference time.

6 EXPERIMENTS

The experiments are organized to test the paper’s main claims: self-distilled supervision can replace real multi-view training data, latent-space reconstruction can scale to many generated views, and the same framework supports both static and dynamic generation.

Table 1: **State-of-the-art comparisons.** Single-image-to-3D results on RealEstate10K, DL3DV, and Tanks-and-Temples show the quantitative effect of self-distilled latent 3DGS generation relative to prior work.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336



Figure 6: **Image-to-3DGS Generation.** Five rendered views from each generated 3DGS scene show that the feed-forward output is an explicit representation, not only a generated image sequence.

6.1 EXPERIMENTAL SETUP

Datasets. To evaluate the data contribution directly, we train the model without any existing multi-view datasets. Instead, we construct the *Lyra* dataset using only our video model as supervision for the 3DGS decoder. The static 3D reconstruction setup contains 59,031 images, and the 4D setup contains 7,378 videos. The data come from diverse text prompts covering indoor and outdoor environments, humans, animals, and both realistic and imaginative content. For each image (3D) or video (4D), we synthesize six camera trajectories, producing 354,186 videos for 3D and 44,268 videos for 4D.

Baselines. We compare against state-of-the-art generative single-image-to-3D reconstruction methods: ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Because *no source code is available* for evaluating these methods on our out-of-distribution evaluation set, our main quantitative comparison relies on the numbers reported in the original papers.

Evaluation. We follow prior work and evaluate single-image-to-3D generation on RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017). The protocol follows Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b). We report standard reconstruction metrics—PSNR, SSIM, and LPIPS—so the gains can be read against established benchmarks.

6.2 MAIN RESULTS

Quantitative results. Tab. 1 shows that the self-distilled framework improves single-image-to-3D reconstruction across all benchmarks and all metrics. This result supports the central claim that a camera-controlled video diffusion teacher can provide supervision strong enough to train an explicit 3DGS student. It also indicates that future improvements in the underlying video generative model should directly transfer to higher-quality reconstructions. Additional quantitative comparisons on the *Lyra* dataset are provided in Appendix C.



Figure 7: **Ablations.** Each variant is evaluated by rendering the same extreme novel viewpoint after image-to-3DGS generation, making the role of each contribution visible; depth visualizations and additional ablations are provided in Appendix C.4.

Qualitative results. Fig. 1 and Fig. 6 visualize the practical form of the contribution: generated 3DGS scenes can be rendered from novel viewpoints, with video results to be released upon acceptance. The renderings show high-quality 3D/4D novel-view content in unseen regions while remaining consistent with the input image or video. Additional qualitative comparisons on the *Lyra* dataset appear in Appendix C.

6.3 ABLATIONS

Tab. 2 isolates which components make the contribution work on out-of-distribution diverse prompts. The ablations compare 3DGS renderings against videos generated by our camera-controlled video diffusion model, so each result measures how well the student preserves the teacher’s generated supervision.

Real data only. This ablation tests whether conventional real multi-view data can replace the proposed self-distilled data source. We train only on RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b), two datasets commonly used in prior work, and observe limited generalization to out-of-distribution scenes.

Self-distillation + real data. Joint training with self-distillation and real data does not improve over self-distillation alone. This supports the contribution that the generated teacher supervision is already diverse and consistent enough to train the reconstruction model.

No depth loss. Removing depth loss weakens the explicit-geometry contribution by allowing flatter reconstructions, and it also slightly reduces image-based metrics. Depth visualizations are provided in Appendix C.4.

No opacity pruning. Keeping all Gaussians instead of pruning by opacity makes the output less compact and slightly lowers visual quality. With pruning, rendering at $H = 704$ and $W = 1280$ takes 18ms, compared with 30ms without pruning, giving a $1.67\times$ speed up.

No LPIPS loss. The LPIPS loss contributes robustness to teacher-student inconsistencies and improves preservation of high-frequency details.

No multi-view fusion. A naive alternative is to generate 3D Gaussians independently for each camera trajectory and then fuse them into one point cloud. The ablation shows the value of learned multi-view fusion: reconstruction blocks let tokens from different trajectories attend to one another, producing substantially better results.

No Mamba-2. Replacing the joint Transformer/Mamba-2 design with Transformer-only blocks slightly lowers quality and substantially reduces efficiency. One forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$ takes 3213ms with the joint blocks, compared with 20922ms with Transformer-only blocks, a $6.5\times$ speedup.

Table 2: **Ablation study on *Lyra* dataset.** Each row isolates a data, loss, or architecture contribution while preserving the same evaluation metrics.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS		OOM	

No latent-based 3DGS. This ablation highlights why the decoder operates in video latent space. Pixel-space methods such as BTimer (Liang et al., 2025b) are restricted to 12 input frames, whereas our latent design can take up to 726 input frames. Moving the 3DGS decoder to pixel space therefore leads to out-of-memory.

7 CONCLUSION

We presented *Lyra*, a 3D and 4D generation framework that turns monocular image or video input into explicit, real-time-renderable 3D Gaussian scenes. The paper’s main contribution is a self-distillation formulation in which a camera-controlled video diffusion model supplies teacher supervision for a latent-space 3DGS student, eliminating the need to collect real multi-view training datasets. The latent decoder enables efficient multi-trajectory fusion, produces 3D Gaussians directly without post-processing or per-scene optimization, and extends to dynamic scenes through time-conditioned 3DGS prediction. The experiments show that these contributions improve generalizability and coverage across diverse static and dynamic scenes.

The main limitation follows from the same contribution structure: the scale and consistency of generated scenes are bounded by the capacity of the camera-controlled video diffusion teacher. Stronger video diffusion models should therefore improve large-scale, consistent scene synthesis within our framework. Another promising direction is to adapt auto-regressive techniques (Chen et al., 2024a) for larger-scale generation, and modelling motion and tracking information inside the reconstruction network, as in concurrent work (Lin et al., 2025), could further improve visual motion quality.

8 ETHICS STATEMENT

Like other generative AI technologies, the method could be misused to create misleading synthetic 3D or 4D content. Our intended contribution, however, is to support simulation, robotics, and embodied AI by making controllable scene generation more scalable and explicit. We expect the most beneficial impact to come from the following capabilities.

- **The ability to enable realistic simulation for embodied agents:** The method provides controllable and physically consistent camera motion within generated 3D and 4D environments, supporting training and evaluation for agents that perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** By removing the need for multi-view capture and expensive per-scene optimization, the framework can produce high-fidelity, customizable scenes at scale for robotics, reinforcement learning, and closed-loop simulation research.
- **Better technical understanding:** As an academic study, the paper clarifies how camera control and spatiotemporal generative models can supervise explicit 3D reconstruction, benefiting research in computer vision, graphics, and embodied AI.

We caution against deceptive or wrongful use of realistic generative outputs, especially when misleading or unverifiable media could be produced. Safeguards such as provenance tracking, dataset documentation, and rigorous evaluation practices should accompany deployment so that generative 3D and 4D models advance science and society in positive ways.

9 REPRODUCIBILITY STATEMENT

To make the contribution reproducible and useful for follow-on research, we will release training and inference code, model weights, and the data used for both 3D and 4D generation upon acceptance.

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1188 A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

1189 This section expands the video-teacher component behind our self-distillation contribution. We
1190 describe the GEN3C (Ren et al., 2025) backbone introduced in Sec. 3.1 and the refinement we use to
1191 make its 3D-consistent supervision more reliable.

1192 **Conservative mask refinement.** This refinement supports the paper’s core claim that camera-
1193 controlled video outputs can supervise explicit 3D reconstruction. A limitation of GEN3C’s forward
1194 warping is background leakage: occluded object parts in the source view (e.g., the lion body in Fig. 8)
1195 become visible from novel viewpoints but are not properly indicated in the disocclusion mask. The
1196 video diffusion model then under-completes the foreground, as shown in Fig. 8 (second and fourth
1197 columns).

1198 We address this failure mode with a conservative masking strategy. Because the geometry of occluded
1199 regions in $\mathbf{P}^{t,v}$ is unknown, we assume such areas may be occupied and derive refined disocclusion
1200 masks accordingly. The teacher can then reason about potentially visible foreground regions that
1201 standard point-based rendering would miss.

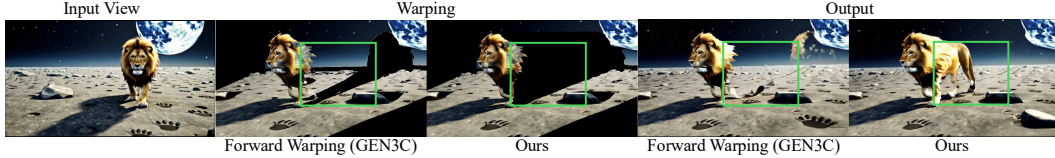
1202 Concretely, we construct a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting
1203 spatially adjacent pixels, similar to Hu et al. (2021):

$$1204 \mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

1205 where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. The mesh acts as a conservative boundary
1206 between visible and invisible regions. For each rendered pixel, we compare the standard point-based
1207 depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained via ray-surface intersection:

$$1208 \mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

1209 where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask from forward warping and ϵ is a small tolerance
1210 factor. When the mesh surface is closer than the original points, we mask the region as a potential
1211 foreground disocclusion. The resulting teacher guidance is more reliable under large viewpoint
1212 changes, as shown in Fig. 8 (third and fifth columns).



1220 Figure 8: **Rendering function comparison.** Our model builds on GEN3C (Ren et al., 2025), which
1221 uses forward-warped images for camera-control conditioning. Comparing forward warping (left)
1222 with our conservative rendering function (right) shows why masking matters: the highlighted region
1223 is incorrectly filled with background pixels under forward warping but is properly masked by our
1224 renderer, enabling correct completion.

1232 B MODEL DETAILS

1233 **Progressive training.** Progressive training makes the latent reconstruction contribution computationally
1234 feasible at full scale. Directly training the 3DGS decoder on 6 trajectories of 121 frames
1235 at 704×1280 is expensive, so we increase the training setting in stages, as shown in Tab. 3. We
1236 then initialize the dynamic model from the final static pre-trained model. Training takes 6 days on 8
1237 NVIDIA A100 80 GB GPUs, and we use gsplat (Ye et al., 2025) as the 3DGS implementation.

1238 **Subsampling Gaussians.** Subsampling is another scaling choice that preserves the explicit 3D output
1239 while keeping it tractable. Pixel-aligned Gaussian generation scales linearly with spatial and temporal
1240 resolution; representing 726 frames at 704×1280 resolution would produce 654,213,120 per-pixel
1241

Table 3: **Progressive training setup.** Static and dynamic stages scale resolution, sequence length, view count, and supervision views for latent-space 3DGS distillation.

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1–6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

Gaussians. We therefore generate one 3D Gaussian per 8×8 spatial neighborhood, reducing the count by a factor of 64 to 10,222,080. Opacity-based pruning further reduces the representation to 2,044,416 Gaussians, making the final scene more compact.

Joint Transformer-Mamba blocks. The reconstruction blocks are chosen to balance fusion quality and efficiency. We use the Long-LRM (Ziwen et al., 2024) design, where one block consists of one Transformer layer followed by seven Mamba-2 layers, and repeat the block twice to obtain 16 layers with 512 hidden dimensions. This network is substantially smaller than Long-LRM (Ziwen et al., 2024) or Wonderland (Liang et al., 2025a), which use 24 layers with 1024 hidden dimensions. Pure Transformers can be powerful, but they are significantly slower in training and inference than Mamba2. Because both configurations were trained under the same budget, we hypothesize that transformer-only blocks would require many more iterations to match the rendering quality.

Camera encoding. Camera encoding aligns the geometry signal with the latent reconstruction space. Following prior work (Kant et al., 2024; Zhang et al., 2024e), we use Plücker embeddings computed at full spatial and temporal resolution and project them into the video latent space with the pre-trained RGB encoder. Because the RGB encoder expects 3 input channels while Plücker embeddings are 6-dimensional, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins, then concatenate the resulting latent encodings along the channel dimension.

Time encoding. Time encoding is the mechanism that carries the static contribution into dynamic scenes. To make the 1-dimensional time input compatible with the 3-channel RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to the original time value along the channel dimension. We replicate these values across spatial dimensions before encoding time into the video latent space, allowing source and target times to be independently represented in compressed latent form.

Dynamic data augmentation. Fig. 9 visualizes the augmentation that makes dynamic self-distillation robust. The procedure creates paired supervision views that observe the same motion state from different extreme viewpoints, reducing the supervision imbalance across time.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We include BTimer (Liang et al., 2025b) to clarify the advantage of latent-space reconstruction over a pixel-space feed-forward baseline for joint 3D and 4D reconstruction from posed images or videos. We contacted the BTimer authors to run experiments on the same scenes as *Lyra*. Because BTimer is purely regression-based, we integrate it with our GEN3C (Ren et al., 2025) video diffusion backbone: we use the same camera trajectories as our model and provide RGB-decoded GEN3C videos as BTimer input. BTimer requires 12 pixel-space input images, whereas directly using all 726 high-resolution GEN3C frames would exceed memory, as discussed in the main paper. We therefore uniformly subsample GEN3C frames. For static evaluation, the 12 frames evenly cover the available viewpoints. For dynamic evaluation, 11 frames span the viewpoint range and one additional bullet-time frame corresponds to the reconstructed motion, ensuring the target motion state is present

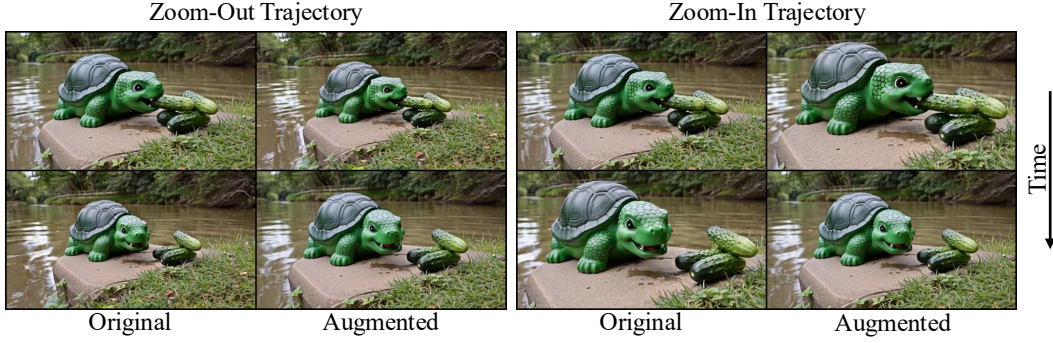


Figure 9: **Dynamic data augmentation videos.** Motion-reversed supervision ensures that each timestep is observed with broader spatial coverage, preventing low-opacity artifacts at early timesteps. We show zoom-out and zoom-in trajectory examples and their corresponding augmented videos, whose camera motion is flipped.



Figure 10: **Image-to-3DGS Comparison.** Five-view renderings compare our generated 3DGS scenes with BTimer (GEN3C). The results show the visible effect of self-distilled latent reconstruction: fewer artifacts and higher fidelity.

as a reference. We denote this integrated baseline as BTimer (GEN3C), i.e., BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 and Tab. 4a show that the self-distilled latent decoder produces higher-quality results with fewer artifacts than BTimer (GEN3C).

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare with BTimer (GEN3C), defined in Sec. C.1, on 100 out-of-distribution videos from the dynamic *Lyra* dataset. We crop videos to 512×512 for fairness because BTimer was mainly trained at that resolution. Tab. 4b shows that the time-conditioned version of our framework significantly outperforms BTimer (GEN3C).

Table 4: **Quantitative results on static and dynamic *Lyra* datasets.** The comparison reports how the same self-distilled framework performs for both feed-forward 3D and time-conditioned 4D generation.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) *Static Lyra* dataset. (b) *Dynamic Lyra* dataset.

C.4 DEPTH VISUALIZATION

Fig. 11 visualizes depth from generated 3DGS scenes and supports the role of depth supervision in the method: it prevents flat geometry without sacrificing visual quality.

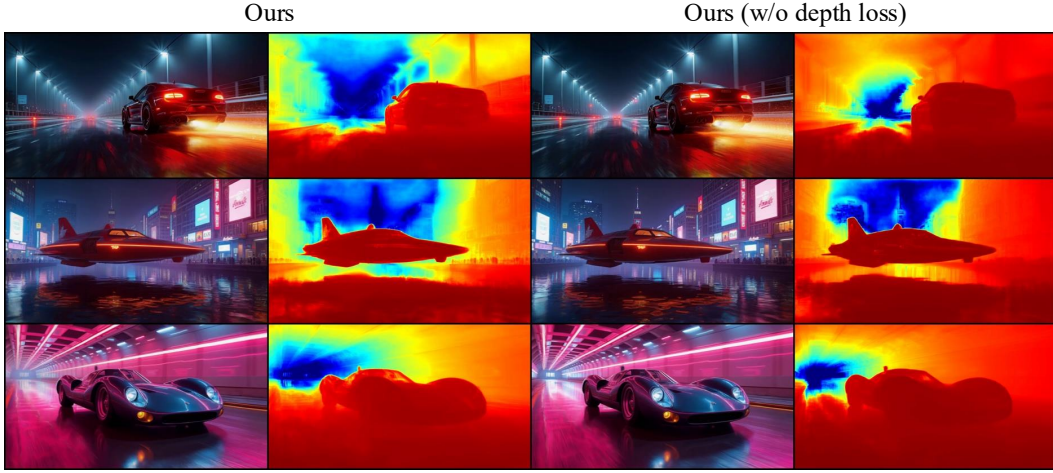


Figure 11: **Depth loss ablation.** Renderings and depth maps show the geometric contribution of depth supervision: it prevents flat 3DGS geometry while preserving visual quality.

D ROBOT SIMULATION

A key motivation for explicit 3D generation is downstream simulation for autonomous agents. We demonstrate this use case with an end-to-end pipeline: generate 3D scenes from text with our model, export the 3D Gaussians as a .ply file, and convert the .ply files into .usdz format. We use 3DGUT (Wu et al., 2025a) and its export function to create the .usdz files. The resulting .usdz scene can be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment, where AI-based robots can be trained and tested under diverse conditions. Early demo results will be released upon acceptance, and Fig. 12 shows a screenshot of the interface.

E RELATED WORK

This section adds related-work context for the contribution claims in the main paper. Fig. 13 summarizes how our approach differs from CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b).

3D generation. The broader 3D generation literature establishes the need for scalable, explicit scene synthesis. Early approaches focused mainly on single object categories and extended GANs into 3D using neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-EI et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). CLIP-based supervision (Radford et al., 2021) broadened the field toward text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). Diffusion-based methods further improved quality by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole

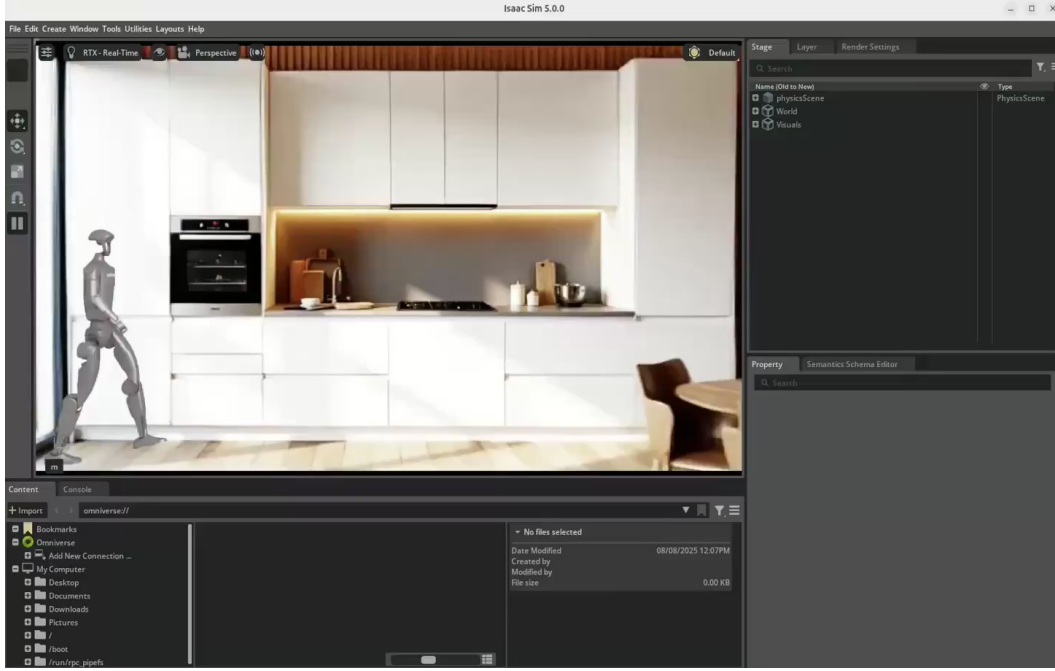


Figure 12: **Robot Simulation.** This example illustrates the downstream utility of the explicit 3DGS output: a generated scene from our method is used as input to the Isaac Sim 5.0 robot simulation framework.

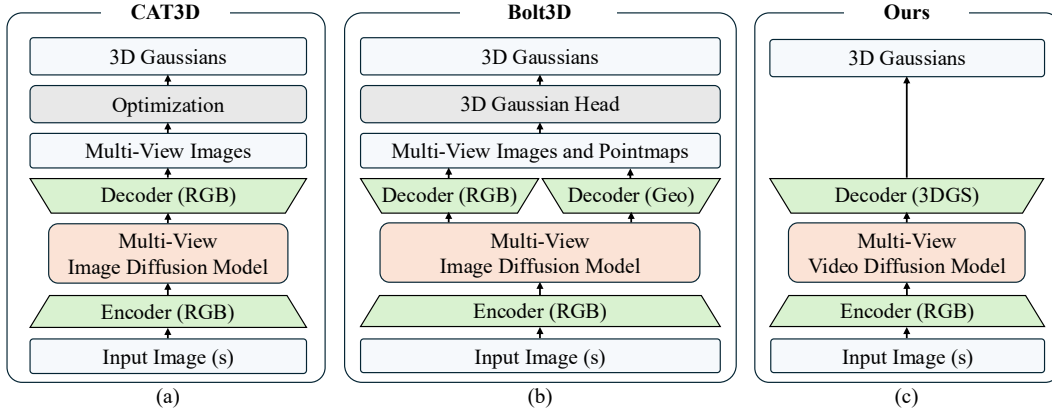


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) generates novel-view images with a multi-view image diffusion model and then reconstructs them into 3D Gaussians through optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps with a geometry autoencoder and predicts a 3D scene using a feed-forward 3D Gaussian head. Our contribution differs by using a pre-trained camera-controlled video diffusion model as supervision and *directly* decoding multi-view video latents into 3D Gaussians.

et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

Many methods improve structural coherence by enforcing multi-view consistency across generated perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), while others synthesize scenes through iterative inpainting (Höllein et al., 2023; Shriram et al., 2024). A related line lifts 2D images into NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based

representations coupled with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024). Our contribution differs by amortizing explicit scene reconstruction through self-distillation rather than optimizing each scene separately.

Feed-forward 3D models. Feed-forward methods are closest to our goal of fast explicit generation because they directly predict 3D content from images or text (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). Most, however, produce static 3D scenes, and specialized methods target categories such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) addresses real scenes but cannot handle diverse generated scenes or large viewpoint changes. Our contribution combines feed-forward inference with self-distilled video supervision and extends the same structure to dynamic generation.

4D generation. 4D generation provides the dynamic counterpart to our extension. Many methods synthesize dynamic 3D scenes from text prompts or images, and since the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent work has improved both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

Beyond text conditioning, many systems lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Other work adds physics priors (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), improves controllability with templates (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b), or studies compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024). Our dynamic contribution is complementary: it distills monocular video supervision into a time-conditioned explicit 3DGS representation.

Another strand extends 3D GAN frameworks to 4D by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). These methods are often limited by small-scale, single-category datasets, which constrains generalization, and many remain object-centric with limited background modeling. Our framework instead leverages the photorealism and diversity of recent video generation models as teacher supervision.

Camera-conditioned video models. Camera-conditioned video models supply the controllable teacher used in our framework. MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Later methods (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) use Plücker coordinates for camera representation, while others follow MotionCtrl by using flattened poses (Wang et al., 2025b). Training-free methods (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enable camera motion control without additional trainable parameters. Most of these advances rely on U-Net architectures, and other approaches explore low-rank camera control (Zhang et al., 2025b).

Recent U-Net-based methods continue to improve camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while diffusion-transformer approaches are emerging. Cheong et al. (Cheong et al., 2024) propose a transformer-based framework, though its scene and visual quality remain limited. DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers but depends on pre-defined, non-continuous camera trajectories. Other works (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) study pose and geometry estimation with a video DiT. CAT4D (Wu et al., 2025b) fine-tunes a multi-view backbone into a multi-view video diffusion model, and large-scale synthetic data improves transformer-based

camera-controlled generation in SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a). 4DiM (Watson et al., 2024) trains a space-time diffusion model from scratch for novel view synthesis from a single input image. Other work retargets camera motion in videos with underlying motion (Seo et al., 2025; Ma et al., 2025), and many methods condition camera-controlled generation with a spatiotemporal cache such as point clouds or optical flow (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025). Our contribution uses this progress as supervision for an explicit feed-forward 3D/4D output rather than treating the video as the final product.